When the biomarker is also prognostic: bias in the biomarker-treatment interaction test of aggregated N-of-1 trials and two remedies

First draft, assembled from the material that left paper 02

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Abstract

Background. The biomarker-treatment interaction test of an aggregated N-of-1 trial is built for a predictive biomarker. A biomarker can also be prognostic. It can then be associated with the natural history, the placebo response or the baseline level. We asked when such an association biases the interaction test, and what removes the bias.

Methods. We simulated the Hybrid design under the extended data-generating process (DGP) of the companion paper, with the parameter values published by Hendrickson et al. Five prognostic pathways were examined. Three are correlations of the biomarker with a component. Two are mean shifts of the natural-history component, one constant and one growing over the trial. We compared analyses with and without the design-expectancy terms η_t and $B_i^c \eta_t$.

Results. Under a correlation with the placebo-response component of 0.45, the analyses without expectancy terms rejected a true null at 0.092 (G9) at $N = 280$. With the terms, the rate was 0.060 (F1). The two-MCSE band at 1,000 replicates was $[0.036, 0.064]$. Under a natural-history shift that grows over the trial, G9 rejected at 0.073 and F1 at 0.052 at $N = 70$. Without a prognostic effect, F1A0 reached 98.2 percent of the power of G10 in the Hybrid cells. The power loss once attributed to a strong prognostic correlation came from repaired covariance matrices. Under the current stop on infeasible matrices, all 10 of those cells were refused.

Conclusions. For a biomarker that may be prognostic, add η_t and $B_i^c \eta_t$ to the recommended analysis of the companion paper. A biomarker that may predict the rate of natural-history change needs a biomarker-by-time term. That remedy has not yet been simulated.

Notation and glossary

Symbols follow `analysis/report/NOTATION.md`. Two symbols are not yet listed there and are marked *proposed*.

Symbol	Meaning
B_i, B_i^c	biomarker of participant i; centered at the trial's baseline mean
z_i	standardized biomarker
w_t	calendar week of visit t
c_{bm}	on-drug biomarker-response correlation, the interaction under the correlation encoding
$c_{bm,PB}$, $c_{bm,TV}$, $c_{bm,BL}$	prognostic correlations of the biomarker with the placebo-response component, the natural-history component and the baseline level
b_{TV} (*proposed*)	constant natural-history mean shift per SD of the biomarker
b_{TV}^{tr} (*proposed*)	coefficient of the trend pathway: a shift of $b_{TV}^{\mathrm{tr}}\, \sigma_{TV}\, z_i\, g(w_t)$, with g the natural-history mean course scaled to peak 1
σ_{BR}, σ_{PB}, σ_{TV}	SDs of the drug-response, placebo-response and natural-history components
$t_{1/2}$, $t_{1/2}^a$	carryover half-life of the DGP; half-life assumed by the analysis
$D_{bc,it}$	exposure-weighted drug indicator built at $t_{1/2}^a$
η_t	design expectancy: 1 open label, 0.5 blinded, 0 at baseline
$\beta_{bm:D}$	interaction coefficient, the target of the test

Symbol	Meaning
β_η, $\beta_{bm:\eta}$	coefficients of η_t and $B_i^c\eta_t$
κ	mean model standard error divided by the empirical SD of the estimates
$\hat{\pi}$, MCSE	estimated rejection rate and its Monte Carlo standard error

- **Correlation encoding.** The interaction is a correlation between the biomarker and the drug response. It is present on drug and decays after discontinuation at the carryover half-life. Code labels it `mvn`.
- **Original DGP; extended DGP.** The original DGP is the code published by Hendrickson et al. at commit `58b32a9` (1). The extended DGP is the one recommended in paper 01 (2). It has a decaying biomarker-response correlation, the same first-order autoregressive correlation for every component with a constant between-component correlation at every lag ($\rho = 0.7$, $c_1 = 0.2$), carryover that decays from the last on-drug mean, and a stop on correlation matrices that are not positive definite. Code and data label it `refined`.
- **Published parameter set; simcore2026 set.** The published set is the component means and SDs of Hendrickson et al. The simcore2026 set is an earlier set used by the paper 02 drivers before 2026-11-03. Runs under it are labeled historical in this draft.
- **Predictive biomarker.** One that changes the drug response ($c_{bm} \neq 0$). **Prognostic biomarker:** one associated with the outcome through another channel. A biomarker can be both.
- **Prognostic pathway.** One channel of prognostic association. The stored data labels are `prog-BL` ($c_{bm,BL}$), `prog-TV-const` ($c_{bm,TV}$), `prog-PB` ($c_{bm,PB}$), `prog-TV-grow` (b_{TV}, constant despite its label) and the block T trend pathway (b_{TV}^{tr}).
- **Feasibility bound.** The largest value of one correlation, given the others, for which the joint correlation matrix is positive definite on every path of the design.
- **Silent repair; repaired cell.** Before 2026-10-26 the drivers replaced a matrix that was not positive definite with a nearby one that was, without warning. A repaired cell is a cell simulated under such a matrix.
- **G-family.** Analyses with no expectancy terms. **G8:** $D_{bc,it}$ at the true half-life with CR2, an oracle benchmark and not a candidate analysis. **G9:** $t_{1/2}^a$ chosen by AIC on the model with the interaction, with CR2. **G10:** $t_{1/2}^a$ chosen by AIC on the model without the interaction, the interaction model refitted, with CR2. G10 is the recommended analysis of paper 02 (3). G1 and G6 are the binary-indicator analyses, model-based and with CR2. Stored codes: `E2cr2`, `E7cr2`, `E7null_cr2`, `E1`, `E1cr2`.
- **F-family.** The G3 formula plus η_t and $B_i^c\eta_t$, with CR2. **F1:** $t_{1/2}^a$ fixed at the half-life policy value of 0.2 week. **F1A:** chosen by AIC with the interaction in the model. **F1A0:** chosen by AIC without it. **F1A0B:** F1A0 with a single-resample parametric-bootstrap calibration that repeats the selection.
- **CR2.** The Bell-McCaffrey bias-reduced cluster-robust standard error with Satterthwaite degrees of freedom (4,5).

- **Two-MCSE band.** The interval $0.05 \pm 2\sqrt{0.05 \times 0.95/n}$ at n replicates.
- **Designs.** Hybrid: open-label titration, then blinded discontinuation, then a brief crossover. OL+BDC: open-label titration then blinded discontinuation. CO: a two-period crossover.
- **Single-resample bootstrap.** One bootstrap resample per Monte Carlo replicate, with the critical value taken from the pooled resamples of the cell.

1 Introduction

An aggregated N-of-1 trial gives each participant repeated periods on and off drug. Pooling the participants allows a test of whether a baseline biomarker modifies the drug effect. That test targets a predictive biomarker. The interaction test in a parallel-group trial already needs a large sample (6). The N-of-1 design reduces that cost by comparing each participant with themselves (7,8).

A biomarker can also be prognostic. It may track the severity at baseline, the natural course of the illness, or the response to placebo. In a design whose expectancy varies between open-label and blinded phases, the on-drug and off-drug visits differ in more than drug exposure. A prognostic association that differs between those visits can then project onto the interaction term. The test would then report moderation of the drug effect where there is none.

The companion paper on carryover (3) studies a predictive-only biomarker. It recommends G10. That analysis uses an exposure-weighted drug indicator, chooses its half-life by AIC on the model without the interaction, and tests with CR2. This paper asks what changes when the biomarker is also prognostic. It examines five pathways. It compares the G-family with the F-family, which adds the design-expectancy terms η_t and $B_i^c\eta_t$. It also reports a finding about the simulation itself. An apparent loss of power under a strong prognostic correlation came from the silent repair of infeasible covariance matrices. It was not a property of any analysis.

The data-generating facts are established in paper 01 (2). A prognostic correlation shares one feasibility condition with c_{bm}. The extended DGP refuses a matrix that is not positive definite. This paper cites those results and reports only their consequences for the analysis. The studies are exploratory.

2 Methods

2.1 Data-generating process

All published-set cells use the extended DGP under the correlation encoding, on the Hybrid design, with $t_{1/2} = 0.5$ week unless stated. The component means and SDs are those of Hendrickson et al. Table 1 reads them from the stored run metadata. The drug-response SD is $\sigma_{BR} = 8$. The true interaction at c_{bm} is therefore $-c_{bm}\sigma_{BR}$ per SD of the biomarker: -3.60

Table 1: Parameter values of the published-set runs, read from the stored run metadata (`meta.rds`, field `params`). For the response components the mean column is the maximum m of the modified Gompertz curve.

Component	Maximum or mean	SD	Rate (per week)	Displacement
Natural history (TV)	6.51	10	0.35	5
Placebo response (PB)	6.51	10	0.35	5
Drug response (BR)	10.99	8	0.42	5
Biomarker	124.3	15.4	–	–
Baseline level	83.1	18.5	–	–

at $c_{bm} = 0.45$ and -2.40 at $c_{bm} = 0.30$. The components are reductions in symptom severity, so an increase in any component lowers the outcome. Interaction coefficients are negative.

2.2 Prognostic pathways

A biomarker may be associated with the outcome through channels other than the drug response. Three pathways are entries in the biomarker row of the joint correlation matrix. Each sets one correlation with the others at zero. The first is a correlation with the baseline level, $c_{bm,BL}$. The second is a correlation with the natural-history component at every visit, $c_{bm,TV}$. The third is a correlation with the placebo-response component at every visit, $c_{bm,PB}$. Their strengths are bounded by feasibility (Section 3.1).

Two further pathways are mean shifts. They are not bounded by the correlation matrix. The constant shift adds $b_{TV}\,\sigma_{TV}\,z_i$ to the natural-history mean at every visit. It is stored as `prog-TV-grow` although it does not grow. It is absorbed exactly by the biomarker main effect of every analysis here (Appendix A). It serves only as a check of that absorption. The trend pathway adds $b^{\text{tr}}_{TV}\,\sigma_{TV}\,z_i\,g(w_t)$, where g is the natural-history mean course scaled to peak 1. It describes a biomarker that predicts the rate of natural-history change. A model with no biomarker-by-time term cannot absorb it.

2.3 Analyses

All analyses fit a linear mixed model with a participant random intercept and continuous-time AR(1) residual correlation (`corCAR1`). The G-family model is

$$Y_{it} = \beta_0 + \beta_B B^c_i + \beta_T w_t + \beta_D D_{bc,it} + \beta_{bm:D} B^c_i D_{bc,it} + u_i + \epsilon_{it}.$$

$D_{bc,it}$ is one on drug, $2^{-t_{sd}/t^a_{1/2}}$ off drug after first exposure, and zero before first exposure. G8 sets $t^a_{1/2} = t_{1/2}$. It is an oracle benchmark and not a candidate analysis. G9 chooses $t^a_{1/2}$ by maximum-likelihood AIC over $\{0.25, 0.5, 1, 2\}$ weeks with the interaction in the model (9). G10 makes the same choice on the model without the interaction and refits the interaction model by REML. The selection of G9 depends on the coefficient under test. That is a post-selection inference risk (10,11). All three test $\beta_{bm:D}$ with CR2 and Satterthwaite degrees of freedom.

Table 2: Stored runs read by this draft. The parameter set is the value recorded in each run's metadata. Runs made before the parameter switch of 2026-11-03 record none and used the simcore2026 set (docs/74). Two analytic files of the historical folder, `pd-check.rds` and `coupling-ceilings.rds`, are read as well. The first depends on the simcore2026 SDs. The second depends on the correlation matrix only.

Run	Folder under carryover-sensitivity/	Parameter set recorded	Date	Status
Prognostic stress test	`published/g9-prognostic-stress`	published	2026-11-04	published set
Feasible grid and large N	`published/g9-gap-fill/C`	published	2026-11-05	published set
Trend pathway (block T)	`published/referee-variants/block-T`	published	2026-11-04	published set
Hybrid nulls (block N)	`published/referee-variants/block-N`	published	2026-11-04	published set
Hybrid alternatives (block A)	`published/referee-variants/block-A`	published	2026-11-04	published set
Original 30-cell grid	`published/tier1-g-vs-f-e0`	published	2026-11-04	published set
Scenario cells	`published/g-vs-f-comparison`	published	2026-11-04	published set
Prognostic stress test	`g9-prognostic-stress`	not recorded	2026-10-26	historical, repaired
Random-slope variants	`g9-random-effects`	not recorded	2026-10-26	historical, repaired

The F-family adds the design-expectancy terms:

$$Y_{it} = ... + \beta_\eta \eta_t + \beta_{bm:\eta} B_i^c \eta_t.$$

The regressors are `De` and `bmc:De`. The placebo-response amplitude scales with η_t, and η_t differs between on-drug and off-drug visits in the Hybrid design. A biomarker correlated with the placebo response therefore acquires an on/off slope difference by construction. The term $B_i^c \eta_t$ absorbs it. F1 fixes $t_{1/2}^a$ at 0.2 week. F1A selects it by AIC with the interaction. F1A0 selects it without the interaction, as G10 does. F1A0B calibrates F1A0 by a single-resample parametric bootstrap under the null that repeats the selection. Its candidate grid is $\{0, 0.2, 0.5, 1, 2\}$ weeks. η_t is constant within every path of the crossover design. The F-family is therefore not estimable under CO. F1A0 and G10 differ only in the expectancy terms. That is the property that lets papers 02 and 15 give one recommendation (Section 4.2).

2.4 Performance measures

We report the rejection rate $\hat{\pi}$ at $\alpha = 0.05$, its MCSE, the mean interaction estimate, and κ (12). Under the null $\hat{\pi}$ is the Type I error. A cell's rejection rate is compared with its two-MCSE band. That band is a check on gross departures and not a demonstration of nominal size. Within a job, cells at the same design and N share random-number streams, and every analysis is fitted to the same trials. Contrasts between analyses are paired, and their standard errors come from the per-replicate rejection indicators. Comparisons across jobs are unpaired. Results from different simulation hosts are not draw-for-draw identical.

2.5 Stored runs and their parameter sets

Table 2 lists every stored run this draft reads. The published-set runs are the reruns of `docs/74`. The historical runs used the simcore2026 parameter set and silent repair. They are kept only as the exhibit of Section 3.8 (`docs/74`, decision D5). No historical number enters a result about an analysis.

Table 3: Rejection rate under each prognostic pathway at strength 0.3, published parameter set. Hybrid, $t_{1/2} = 0.5$ week, 500 replicates per cell, the same trials for every analysis. Rows with $c_{bm} = 0$ are Type I error (band $[0.031, 0.069]$). G8 is the oracle benchmark.

Pathway	c_{bm}	N	G8	G9	G10	F1	F1A0	F1A0B
$c_{bm,PB}$	0.00	35	0.040	0.044	0.038	0.032	0.034	0.032
	0.00	70	0.060	0.058	0.058	0.042	0.038	0.032
	0.45	35	0.484	0.500	0.468	0.380	0.379	0.415
	0.45	70	0.786	0.792	0.780	0.658	0.678	0.646
$c_{bm,TV}$	0.00	35	0.038	0.042	0.038	0.034	0.032	0.034
	0.00	70	0.056	0.050	0.048	0.042	0.036	0.030
	0.45	35	0.418	0.431	0.402	0.371	0.377	0.405
	0.45	70	0.714	0.716	0.706	0.662	0.668	0.654
$c_{bm,BL}$	0.00	35	0.040	0.044	0.038	0.032	0.032	0.034
	0.00	70	0.056	0.054	0.048	0.042	0.038	0.032
	0.45	35	0.426	0.448	0.414	0.381	0.377	0.411
	0.45	70	0.720	0.730	0.712	0.682	0.692	0.648
b_{TV} (constant)	0.00	35	0.038	0.042	0.038	0.030	0.032	0.034
	0.00	70	0.056	0.054	0.048	0.042	0.038	0.032
	0.45	35	0.409	0.424	0.393	0.363	0.369	0.403
	0.45	70	0.720	0.724	0.710	0.694	0.698	0.680

3 Results

3.1 Feasibility of a prognostic correlation

Paper 01 derives the joint feasibility condition and tabulates the bounds for every design (2). A prognostic correlation occupies the same biomarker row of the correlation matrix as c_{bm}. The two share one condition, a quadratic form in the correlations that must be below one. We do not repeat that table. We quote only the bounds at this paper's configuration: Hybrid, $t_{1/2} = 0.5$ week, $\rho = 0.7$, $c_1 = 0.2$. The bounds depend on the correlation matrix only, not on the component SDs. They therefore hold for both parameter sets.

Alone, c_{bm} is feasible up to 0.518 and $c_{bm,PB}$ or $c_{bm,TV}$ up to 0.513. Given $c_{bm} = 0.30$ the bound on $c_{bm,PB}$ or $c_{bm,TV}$ is 0.451 and on $c_{bm,BL}$ is 0.815. Given $c_{bm} = 0.45$ they are 0.306 and 0.496. No c_{bm} is feasible with $c_{bm,PB}$ or $c_{bm,TV}$ at 0.6. Strength 0.3 therefore lies just inside the bound at $c_{bm} = 0.45$. Strength 0.6 is infeasible at every c_{bm} on those two pathways. The feasible grid of Section 3.3 was chosen from these bounds.

3.2 Correlation pathways at strength 0.3

We crossed the three correlation pathways and the constant shift with $c_{bm} \in \{0, 0.45\}$ and $N \in \{35, 70\}$, at 500 replicates per cell. Table 3 gives the rejection rates. The two-MCSE band is $[0.031, 0.069]$.

The G9 Type I error was 0.042 to 0.058 across the six null cells of the three correlation pathways. G10 gave 0.038 to 0.058 and F1A0B 0.030 to 0.034. Every G9 and G10 null rate was inside the band. The G9 power at $c_{bm} = 0.45$ was 0.424 to 0.792. The no-confound reference from block A of the referee-variant runs is 0.442 at $N = 35$ and 0.728 at $N = 70$.

Table 4: Block N, published parameter set: null rejection rate (MCSE) under the placebo-response pathway at $c_{bm,PB} = 0.3$, Hybrid, 5,000 replicates per cell, the same trials for every analysis, with the paired difference from G8 and its paired SE. Band $[0.044, 0.056]$.

Cell	Method	Rejection	Diff. vs G8 (SE)	κ
$t_{1/2} = 0.5$, $N = 35$	G8	0.0566 (0.0033)	–	0.95
	G9	0.0653 (0.0035)	+0.0086 (0.0017)	0.88
	G10	0.0573 (0.0033)	+0.0006 (0.0016)	0.94
	F1	0.0541 (0.0032)	-0.0028 (0.0027)	0.96
$t_{1/2} = 0.5$, $N = 70$	G8	0.0530 (0.0032)	–	1.00
	G9	0.0586 (0.0033)	+0.0056 (0.0016)	0.96
	G10	0.0538 (0.0032)	+0.0008 (0.0015)	1.00
	F1	0.0474 (0.0030)	-0.0056 (0.0025)	1.00
$t_{1/2} = 1.0$, $N = 35$	G8	0.0469 (0.0030)	–	0.97
	G9	0.0601 (0.0034)	+0.0130 (0.0021)	0.88
	G10	0.0457 (0.0030)	-0.0012 (0.0019)	0.95
	F1	0.0411 (0.0028)	-0.0060 (0.0030)	0.98
$t_{1/2} = 1.0$, $N = 70$	G8	0.0606 (0.0034)	–	0.95
	G9	0.0740 (0.0037)	+0.0134 (0.0022)	0.89
	G10	0.0622 (0.0034)	+0.0016 (0.0021)	0.93
	F1	0.0534 (0.0032)	-0.0072 (0.0033)	0.96

The largest difference from it was +0.064, under the placebo-response pathway. Section 3.3 shows that this excess is bias in the estimate. That comparison is unpaired, against 1,000-replicate cells of another job. F1A0B differed from G10 in power by -0.134 to +0.010 on the same trials. The constant shift gave the same rejection at strengths 0.3 and 0.6 in every analysis. In the scenario job, the constant-shift null cell differs from the unshifted Hybrid null cell by at most 0.0000 in any analysis. That is the absorption of Appendix A.

A longer run at the placebo-response pathway sharpens the size comparison. Block N of the referee-variant runs fitted G8, G9, G10 and F1 to 5,000 replicates in each of 4 Hybrid null cells at $c_{bm,PB} = 0.3$ (Table 4). The band is $[0.044, 0.056]$. The paired difference from G8 was +0.0056 to +0.0134 for G9, -0.0012 to +0.0016 for G10 and -0.0072 to -0.0028 for F1. G9 was above G8 in every cell, by at least 3.5 paired standard errors. G10 was within two paired standard errors of G8 in every cell. The number of cells above the band was 2 for G8, 4 for G9, 2 for G10 and 0 for F1.

3.3 The feasible prognostic grid

The supplementary study ran a grid of correlations chosen inside the feasibility bounds. At $c_{bm} = 0.30$ it set $c_{bm,PB}$ and $c_{bm,TV}$ at 0.40 and 0.45, and $c_{bm,BL}$ at 0.6 and 0.8. It also ran $c_{bm,BL} = 0.45$ at $c_{bm} = 0.45$. Each cell was run with and without a true interaction, at $N = 70$ and 500 replicates. The strongest $c_{bm,PB}$ lies 0.001 below its bound at $c_{bm} = 0.30$. The strongest $c_{bm,BL}$ lies 0.015 below its bound. Table 5 gives the rejection rates and Table 6 the mean estimates.

The G9 Type I error was 0.052 to 0.066 across the 7 null cells, inside the band in every one. G10 gave 0.048 to 0.062 and F1 0.054 to 0.060. The rejection rates hide a bias that the mean estimates show. Under the placebo-response pathway with no true interaction, the mean G9 estimate was -0.54 to -0.49. G10 gave -0.52 to -0.47. F1 gave -0.09. Under the other two

Table 5: Rejection rate on the feasible prognostic grid, published parameter set. Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, 500 replicates per cell. Rows with $c_{bm} = 0$ are Type I error (band $[0.031, 0.069]$). The others are power.

Cell	G8	G9	G10	F1	F1A0	F1A0B
$c_{bm,PB} = 0.40$, $c_{bm} = 0.00$	0.066	0.056	0.054	0.060	0.058	0.060
$c_{bm,PB} = 0.40$, $c_{bm} = 0.30$	0.564	0.564	0.540	0.400	0.382	0.384
$c_{bm,PB} = 0.45$, $c_{bm} = 0.00$	0.066	0.058	0.054	0.058	0.058	0.058
$c_{bm,PB} = 0.45$, $c_{bm} = 0.30$	0.582	0.580	0.566	0.380	0.390	0.370
$c_{bm,TV} = 0.40$, $c_{bm} = 0.00$	0.040	0.052	0.048	0.054	0.052	0.056
$c_{bm,TV} = 0.40$, $c_{bm} = 0.30$	0.418	0.432	0.412	0.408	0.384	0.386
$c_{bm,TV} = 0.45$, $c_{bm} = 0.00$	0.040	0.054	0.050	0.054	0.050	0.050
$c_{bm,TV} = 0.45$, $c_{bm} = 0.30$	0.439	0.438	0.410	0.406	0.388	0.392
$c_{bm,BL} = 0.45$, $c_{bm} = 0.00$	0.046	0.066	0.062	0.060	0.056	0.056
$c_{bm,BL} = 0.45$, $c_{bm} = 0.45$	0.742	0.746	0.740	0.732	0.726	0.724
$c_{bm,BL} = 0.60$, $c_{bm} = 0.00$	0.046	0.066	0.062	0.060	0.054	0.060
$c_{bm,BL} = 0.60$, $c_{bm} = 0.30$	0.432	0.453	0.427	0.399	0.383	0.391
$c_{bm,BL} = 0.80$, $c_{bm} = 0.00$	0.045	0.061	0.057	0.057	0.055	0.061
$c_{bm,BL} = 0.80$, $c_{bm} = 0.30$	0.444	0.429	0.404	0.404	0.394	0.414

Table 6: Mean interaction estimate on the feasible prognostic grid (same cells as the rejection table). The true value is $-c_{bm}\sigma_{BR}$ per SD of the biomarker.

Cell	Truth	G9	G10	F1	F1A0
$c_{bm,PB} = 0.40$, $c_{bm} = 0.00$	-0.00	-0.49	-0.47	-0.09	-0.06
$c_{bm,PB} = 0.40$, $c_{bm} = 0.30$	-2.40	-2.87	-2.78	-2.17	-2.32
$c_{bm,PB} = 0.45$, $c_{bm} = 0.00$	-0.00	-0.54	-0.52	-0.09	-0.06
$c_{bm,PB} = 0.45$, $c_{bm} = 0.30$	-2.40	-2.94	-2.84	-2.18	-2.34
$c_{bm,TV} = 0.40$, $c_{bm} = 0.00$	-0.00	-0.08	-0.09	-0.09	-0.05
$c_{bm,TV} = 0.40$, $c_{bm} = 0.30$	-2.40	-2.45	-2.39	-2.17	-2.32
$c_{bm,TV} = 0.45$, $c_{bm} = 0.00$	-0.00	-0.08	-0.08	-0.08	-0.04
$c_{bm,TV} = 0.45$, $c_{bm} = 0.30$	-2.40	-2.47	-2.41	-2.19	-2.35
$c_{bm,BL} = 0.45$, $c_{bm} = 0.00$	-0.00	-0.10	-0.09	-0.09	-0.06
$c_{bm,BL} = 0.45$, $c_{bm} = 0.45$	-3.60	-3.64	-3.56	-3.22	-3.47
$c_{bm,BL} = 0.60$, $c_{bm} = 0.00$	-0.00	-0.09	-0.09	-0.09	-0.07
$c_{bm,BL} = 0.60$, $c_{bm} = 0.30$	-2.40	-2.46	-2.39	-2.17	-2.31
$c_{bm,BL} = 0.80$, $c_{bm} = 0.00$	-0.00	-0.09	-0.09	-0.11	-0.08
$c_{bm,BL} = 0.80$, $c_{bm} = 0.30$	-2.40	-2.47	-2.41	-2.18	-2.34

Table 7: Type I error under the placebo-response pathway at larger samples, published parameter set. Hybrid, $t_{1/2} = 0.5$ week, $c_{bm} = 0$, 1,000 replicates at $N = 140$ and 280 (band $[0.036, 0.064]$). The $N = 70$ row is from the feasible grid at 500 replicates.

Cell	G1	G8	G9	G10	F1	F1A0	F1A0B
$c_{bm,PB} = 0.30$, $N = 140$	0.048	0.060	0.062	0.058	0.052	0.050	0.041
$c_{bm,PB} = 0.30$, $N = 280$	0.055	0.065	0.067	0.066	0.056	0.053	0.050
$c_{bm,PB} = 0.45$, $N = 70$	0.046	0.066	0.058	0.054	0.058	0.058	0.058
$c_{bm,PB} = 0.45$, $N = 140$	0.057	0.076	0.077	0.073	0.051	0.049	0.042
$c_{bm,PB} = 0.45$, $N = 280$	0.069	0.094	0.092	0.089	0.060	0.058	0.053

pathways the mean G9 estimate was -0.10 to -0.08. At $c_{bm} = 0.30$ the same bias appears as excess power. G9 reached 0.564 to 0.580 under the placebo-response pathway and 0.429 to 0.453 under the others. Its mean estimate was -2.94 to -2.87 and -2.47 to -2.45 against a true value of -2.40. F1 reached 0.380 to 0.400 under the placebo-response pathway. Its mean estimate was -2.19 to -2.17 across the grid. The F1 estimate is attenuated because its fixed half-life of 0.2 week is below the true 0.5 week (Section 3.7). The excess power of the G-family under the placebo-response pathway is therefore not evidence of better power. It is the same bias as under the null.

3.4 Larger samples under the placebo-response pathway

A bias of a fixed size becomes Type I error inflation as N grows, because the standard error shrinks. The supplementary study ran the placebo-response pathway with no true interaction at $N \in \{140, 280\}$ and $c_{bm,PB} \in \{0.30, 0.45\}$, at 1,000 replicates. The band is $[0.036, 0.064]$. Table 7 gives the result.

At $N = 280$ and $c_{bm,PB} = 0.45$, G9 rejected at 0.092, G10 at 0.089 and G8 at 0.094. The model-based binary-indicator analysis G1 rejected at 0.069. The F-family stayed inside the band: F1 at 0.060, F1A0 at 0.058 and F1A0B at 0.053. On the same trials, the paired difference in rejection of F1 minus G9 was -0.032 (paired SE 0.009). That of F1A0 minus G10 was -0.031 (0.009). The G9 rate rose with N: 0.058 at 70, 0.077 at 140 and 0.092 at 280. Its mean estimate at $N = 280$ was -0.36 against +0.05 for F1. The standard errors were calibrated for both, with κ of 0.96 and 0.97. The excess is therefore bias, not an underestimated standard error. At $c_{bm,PB} = 0.30$ the rates were 0.067 for G9 and 0.056 for F1.

3.5 The trend pathway

Block T of the referee-variant runs simulated a natural-history shift that grows over the trial, with $b_{TV}^{\mathrm{tr}} = 0.3$. It is a mean shift, so the feasibility bound does not limit it. G8, G9 and F1 were fitted at $N = 70$ and 1,000 replicates (Table 8). The band is $[0.036, 0.064]$.

The trend pathway biased the G-family. Under the null G8 rejected at 0.063 and G9 at 0.073. Their mean estimates were +0.43 and +0.42 where the truth is zero. The G8 empirical SE was 1.45. F1 rejected at 0.052, with a mean estimate of +0.24. Paired against G8, F1 differed

Table 8: Block T, published parameter set: rejection rate (MCSE), mean interaction estimate, empirical SE and calibration ratio under the trend pathway ($b_{TV}^{\text{tr}} = 0.3$). Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, 1,000 replicates. The true value is 0 at the null and -3.60 at $c_{bm} = 0.45$. Band at the null $[0.036, 0.064]$.

Cell	Method	Rejection	Mean estimate	Empirical SE	κ
null ($c_{bm} = 0$)	G8	0.063 (0.008)	+0.43	1.45	0.97
	G9	0.073 (0.008)	+0.42	1.46	0.93
	F1	0.052 (0.007)	+0.24	1.27	0.98
$c_{bm} = 0.45$	G8	0.629 (0.015)	-3.20	1.37	0.97
	G9	0.631 (0.015)	-3.12	1.41	0.90
	F1	0.632 (0.015)	-2.88	1.23	0.97

by -0.011 (SE 0.006) and G9 by +0.010 (0.005). The expectancy terms therefore absorb part of the shift but not all of it. At $c_{bm} = 0.45$ the bias is toward zero. G9 power was 0.631 with a mean estimate of -3.12. In the no-trend cell of block A it was 0.728 with -3.47. That comparison is unpaired. The mechanism is the design. In the Hybrid design $D_{bc,it}$ falls with calendar time, so a biomarker that predicts the trend projects onto the interaction term of a model with no biomarker-by-time term.

3.6 The biomarker-by-time remedy: RESULTS PENDING

RESULTS PENDING. The simulation for this section has not been run. This section contains no results. It will be written from the block described in the split plan, Section 3.1, once that block is stored.

The remedy adds a biomarker-by-time term $B_i^c w_t$ to the analysis model, as `bmc:t`. Its coefficient would be written $\beta_{bm:T}$ (*proposed*). A quadratic variant adds $B_i^c w_t^2$. The planned block crosses the trend pathway, present or absent, with a null and an alternative interaction. It fits G8, G9, G10 and F1A0 with and without the term to the same trials. It answers three questions. Does the term remove the trend bias in size and in the mean estimate? What does it cost in power when there is no trend? Does a linear term suffice when the natural-history course is not a straight line? The driver changes it needs are listed in the Reproducibility section.

3.7 Expectancy terms without a prognostic effect

The F-family's cost when the biomarker is predictive only was measured on the original 30-cell grid of paper 02 under the published parameter set (500 replicates per cell). The F-family is not estimable under CO, so only the Hybrid and OL+BDC cells enter. Table 9 gives the rejection rates.

The null rates were 0.044 to 0.062 for G10, 0.054 to 0.056 for F1A0, 0.040 to 0.062 for F1A0B and 0.052 to 0.064 for F1. Over the alternative cells of the Hybrid design, F1A0 reached 98.2 percent of the mean power of G10, F1A0B 97.4 percent and F1 95.5 percent. Under OL+BDC the figures were 93.3, 86.3 and 88.0 percent. The mean paired difference from G10 over the

Table 9: Original 30-cell grid of paper 02, published parameter set, Hybrid and OL+BDC cells: rejection rate with no prognostic effect. 500 replicates per cell, the same trials for every analysis. Rows with $c_{bm} = 0$ are Type I error (band $[0.031, 0.069]$).

Design	c_{bm}	$t_{1/2}$	N	G9	G10	F1	F1A0	F1A0B
Hybrid	0.00	1.0	35	0.078	0.060	0.054	0.056	0.062
	0.00	1.0	70	0.066	0.062	0.052	0.054	0.054
	0.30	1.0	35	0.179	0.155	0.136	0.140	0.154
	0.30	1.0	70	0.327	0.299	0.280	0.280	0.272
	0.45	0.0	35	0.512	0.508	0.486	0.486	0.486
	0.45	0.0	70	0.868	0.868	0.872	0.874	0.874
	0.45	0.5	35	0.394	0.376	0.362	0.374	0.360
	0.45	0.5	70	0.749	0.737	0.718	0.711	0.699
	0.45	1.0	35	0.317	0.277	0.248	0.279	0.287
	0.45	1.0	70	0.597	0.553	0.501	0.561	0.543
OL+BDC	0.00	1.0	35	0.052	0.044	0.064	0.056	0.048
	0.00	1.0	70	0.056	0.050	0.060	0.056	0.040
	0.30	1.0	35	0.114	0.108	0.086	0.106	0.094
	0.30	1.0	70	0.204	0.198	0.150	0.192	0.140
	0.45	0.0	35	0.401	0.390	0.382	0.372	0.384
	0.45	0.0	70	0.721	0.715	0.695	0.682	0.644
	0.45	0.5	35	0.305	0.289	0.220	0.258	0.270
	0.45	0.5	70	0.567	0.563	0.510	0.508	0.454
	0.45	1.0	35	0.202	0.198	0.142	0.178	0.158
	0.45	1.0	70	0.385	0.365	0.300	0.340	0.296

8 Hybrid alternative cells was -0.009 for F1A0 (SE 0.004) and -0.022 for F1 (0.005). Under OL+BDC it was -0.024 (0.005) and -0.043 (0.005). The pooled standard errors treat the cells as independent. Cells at the same N share random-number streams, so they are approximate.

F1 also loses power where its fixed half-life is wrong. In block A at a true half-life of 2 weeks and $c_{bm} = 0.45$, F1 reached 0.163 to 0.279 against 0.192 to 0.359 for G10. Its mean estimate was -1.85 to -1.72 against -3.09 to -2.93 for G10 and a true value of -3.60. The fixed half-life is the cost, not the expectancy terms. F1A0, which selects the half-life, is the fair comparator for the terms themselves.

Table 10: Scenario cells of paper 02, published parameter set, no prognostic effect: rejection rate. Hybrid unless stated, $N = 70$, $t_{1/2} = 0.5$ week unless stated, 500 replicates. In the $t_{1/2} = 2$ week rows the fixed half-life of F1 is a tenth of the true value.

Cell	G9	G10	F1	F1A0	F1A0B
Hybrid, null	0.060	0.052	0.068	0.058	0.056
OL+BDC, null	0.060	0.058	0.048	0.048	0.046
$t_{1/2} = 2$ wk, null	0.078	0.066	0.068	0.058	0.052
Weibull $k = 2$ data, null	0.062	0.058	0.068	0.064	0.054
Hybrid, $c_{bm} = 0.45$	0.754	0.736	0.728	0.722	0.710
Hybrid, $c_{bm} = 0.25$	0.296	0.288	0.290	0.282	0.280
OL+BDC, $c_{bm} = 0.45$	0.576	0.566	0.524	0.532	0.480
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	0.162	0.138	0.104	0.142	0.132
Weibull $k = 2$ data, $c_{bm} = 0.25$	0.378	0.374	0.362	0.362	0.332

Table 11: Historical exhibit (simcore2026 parameter set, silent repair): power at $c_{bm} = 0.45$ at prognostic strengths 0.3 and 0.6, Hybrid, $t_{1/2} = 0.5$ week, 500 replicates. Every strength-0.6 cell here, including $c_{bm,BL} = 0.6$, was simulated under a repaired matrix.

		G9		F1		F1A0B	
Pathway	N	0.3	0.6	0.3	0.6	0.3	0.6
$c_{bm,PB}$	35	0.798	0.597	0.728	0.524	0.741	0.562
	70	0.968	0.853	0.946	0.798	0.940	0.794
$c_{bm,TV}$	35	0.782	0.502	0.720	0.454	0.730	0.482
	70	0.968	0.772	0.952	0.718	0.950	0.702
$c_{bm,BL}$	35	0.782	0.738	0.713	0.676	0.758	0.702
	70	0.976	0.950	0.958	0.929	0.960	0.920

3.8 The repair artifact: a historical exhibit

Historical exhibit. Every number in this section comes from runs under the simcore2026 parameter set with silent repair (Table 2). They are evidence about a repaired DGP. They are not evidence about any analysis under a prognostic biomarker.

The first prognostic stress test ran at strengths 0.3 and 0.6 before the drivers refused infeasible matrices. At strength 0.6 on the placebo-response and natural-history pathways, G9 lost 12 to 26 points of power at $c_{bm} = 0.45$ against the no-confound reference of 0.759 ($N = 35$) and 0.969 ($N = 70$). At strength 0.3 the largest loss was 0.039. Every one of the 14 analyses fitted to the same trials lost power at strength 0.6 (Table 11). That uniformity was the first sign that the loss arose before the analysis.

A direct check of the covariance matrices explains it (Table 12). Each strength-0.6 cell on those two pathways, and $c_{bm,BL} = 0.6$ at $c_{bm} = 0.45$, has a joint matrix that is not positive definite on all 4 Hybrid paths. The repair replaced it with a nearby matrix before any data were drawn. The repair shrank the population on-drug minus off-drug slope of the outcome on the biomarker. That is the quantity the interaction term estimates. With no prognostic correlation it is -3.26. After repair it was -2.38 under the placebo-response pathway and -1.81 under the natural-history pathway. The power loss is the repair removing part of the interaction from the data. No analysis could recover it. At strength 0.3 the matrix is positive definite, with a smallest eigenvalue of 0.010 under the placebo-response pathway.

Before the artifact was found, the loss was read as patient-level variance that a random intercept could not absorb. G9 was refitted with a random slope on time, on expectancy, and on both, in 12 cells at 1,000 replicates. 8 of those cells are repaired strength-0.6 cells. The three variants matched plain G9 to within 0.010 in rejection rate in every cell (Table 13). The largest paired per-replicate difference was 0.008. Non-convergence reached 2.5 percent for the variant with both slopes. The DGP has no patient-specific slope, so there was no variance for a random slope to absorb. The check shows only that the slopes cost nothing there.

The published-set rerun applied the current default, which refuses an infeasible matrix. Of its 12 strength-0.6 cells on the three correlation pathways, 10 were refused and 2 ran. The refused cells are exactly the configurations that the historical check found infeasible. The cells that ran set $c_{bm,BL} = 0.6$ with no true interaction, which is feasible. Under the current code

Table 12: Historical exhibit: positive definiteness of the raw joint covariance matrix on the Hybrid paths, and the population on-drug minus off-drug slope of the outcome on the biomarker before and after the silent repair. Computed from the matrix under the simcore2026 SDs, no simulation (`30-check-prognostic-confound-pd.R`).

Pathway	Strength	c_{bm}	Paths repaired	Min. eigenvalue	Contrast, raw	Contrast, repaired
none	0.0	0.45	0	0.180	-3.26	-3.26
$c_{bm,BL}$	0.3	0.45	0	0.145	-3.26	-3.26
$c_{bm,BL}$	0.6	0.00	0	0.184	+0.00	+0.00
$c_{bm,BL}$	0.6	0.45	4	-0.110	-3.26	-3.00
$c_{bm,PB}$	0.3	0.45	0	0.010	-3.40	-3.40
$c_{bm,PB}$	0.6	0.00	4	-0.240	-0.27	-0.23
$c_{bm,PB}$	0.6	0.45	4	-0.642	-3.53	-2.38
$c_{bm,TV}$	0.3	0.45	0	0.011	-3.26	-3.26
$c_{bm,TV}$	0.6	0.00	4	-0.354	+0.00	+0.00
$c_{bm,TV}$	0.6	0.45	4	-0.853	-3.26	-1.81

Table 13: Historical exhibit: rejection rate of G9 and its three random-slope variants (simcore2026 parameter set, Hybrid, $t_{1/2} = 0.5$ week, 1,000 replicates). The strength-0.6 rows were simulated under repaired matrices.

Cell	G9	+ slope on w_t	+ slope on η_t	+ both
none, $c_{bm} = 0.00$, $N = 35$	0.050	0.050	0.050	0.046
none, $c_{bm} = 0.00$, $N = 70$	0.043	0.044	0.043	0.041
none, $c_{bm} = 0.45$, $N = 35$	0.759	0.762	0.766	0.769
none, $c_{bm} = 0.45$, $N = 70$	0.969	0.970	0.969	0.970
$c_{bm,PB}$ 0.6, $c_{bm} = 0.00$, $N = 35$	0.060	0.060	0.059	0.060
$c_{bm,PB}$ 0.6, $c_{bm} = 0.00$, $N = 70$	0.047	0.046	0.047	0.047
$c_{bm,PB}$ 0.6, $c_{bm} = 0.45$, $N = 35$	0.594	0.593	0.593	0.596
$c_{bm,PB}$ 0.6, $c_{bm} = 0.45$, $N = 70$	0.860	0.859	0.858	0.858
$c_{bm,TV}$ 0.6, $c_{bm} = 0.00$, $N = 35$	0.050	0.053	0.051	0.054
$c_{bm,TV}$ 0.6, $c_{bm} = 0.00$, $N = 70$	0.044	0.044	0.043	0.040
$c_{bm,TV}$ 0.6, $c_{bm} = 0.45$, $N = 35$	0.489	0.488	0.489	0.485
$c_{bm,TV}$ 0.6, $c_{bm} = 0.45$, $N = 70$	0.772	0.772	0.773	0.768

a repaired cell cannot be simulated without an explicit option.

4 Discussion

4.1 What the expectancy terms are for

The placebo-response amplitude scales with η_t. In the Hybrid design η_t differs between on-drug and off-drug visits. A biomarker correlated with the placebo response therefore acquires an on/off slope difference with no true interaction. Under the simcore2026 SDs the population contrast was -0.13 at $c_{bm,PB} = 0.3$ and -0.27 at 0.6, against -3.26 for the interaction at $c_{bm} = 0.45$ (historical, Table 12). The placebo-response SD is 2 in the simcore2026 set and 10 in the published set. The contrast scales with that SD. It has not been recomputed under the published set.

The published-set runs show the consequence. Under the placebo-response pathway the G-family estimate moved away from zero with no interaction present (Section 3.3). The bias was small against the standard error at $N = 70$ and produced no excess rejection there. At $N = 280$ it did (Section 3.4). The F-family, whose $B_i^c \eta_t$ term absorbs the contrast, stayed inside the band at every N examined. The earlier conclusion that the expectancy terms add nothing came from the simcore2026 runs. It does not hold under the published parameter values.

4.2 The joint recommendation

The two companion papers give one recommendation between them.

- For a biomarker believed to be predictive only, use G10: the exposure-weighted indicator with the half-life chosen by AIC on the model without the interaction, tested with CR2 (3).
- Where the biomarker may also be prognostic, add the design-expectancy terms η_t and $B_i^c \eta_t$. This is F1A0, or its bootstrap-calibrated form F1A0B. F1A0 and G10 differ only in those two terms.
- Where the biomarker may predict the rate of natural-history change, also add a biomarker-by-time term. This step is provisional until the block of Section 3.6 is run.

F1, which fixes the half-life at 0.2 week, is not recommended. It loses power wherever that value is wrong (Section 3.7). Two questions remain for F1A0B. Its bootstrap calibration may be unnecessary once the selection excludes the interaction. Its cost against G10 without a prognostic effect is measured here on one grid only.

4.3 Limitations

The published-set results come from one parameter set and one correlation structure ($\rho = 0.7$, $c_1 = 0.2$). Most cells are on the Hybrid design at $t_{1/2} = 0.5$ week. The prognostic correlations

are correlations of a single latent biomarker with the components. The feasibility bound is a property of that parameterization. The large-sample cells have no confound-free counterpart at the same N. The trend pathway was simulated at one coefficient, one sample size and one half-life. Its remedy has not been simulated. The placebo-response contrast has not been recomputed under the published SDs. The random-slope variants were fitted only to historical, partly repaired cells under a DGP without patient-specific slopes. Comparisons with the block A reference are across jobs and unpaired. Cells within a job share random-number streams, so counts of cells inside a band are not independent confirmations.

5 Conclusions

1. A constant prognostic shift is absorbed exactly by the biomarker main effect. It cannot move the interaction estimate.
2. Inside the feasibility bounds, the correlation pathways left the G9 and G10 Type I error inside the band at $N \leq 70$.
3. A biomarker correlated with the placebo response biases the G-family estimate. Under the published parameter values the bias inflated the size of G9 and G10 at $N = 280$. The expectancy terms removed it.
4. A biomarker that predicts the natural-history trend biases every analysis examined. The expectancy terms absorb part of the bias. The biomarker-by-time remedy is pending.
5. The power loss once attributed to a prognostic correlation of 0.6 was produced by the silent repair of infeasible matrices. Under the current default those cells are refused.
6. For a biomarker that may be prognostic, add η_t and $B_i^c \eta_t$ to G10. The result is F1A0.

Declarations

Funding, competing interests, author contributions. To be completed.

Data and code availability. Every number in this draft is read from the stored files of Table 2 by inline code.

6 Reproducibility

The drivers live in `analysis/scripts/carryover-sensitivity/`. The published-set runs were submitted on 2026-11-03 under `docs/74`, wave 1. The prognostic stress test is driver 28. The feasible grid and the large-sample cells are driver 32, block C. The trend, null and alternative blocks are driver 36, blocks T, N and A. The original grid is driver 27 and the scenario cells driver 06. Driver 29, the random-slope variants, was not rerun. Every confounded cell in it is a repaired strength-0.6 cell (`docs/74`, decision D5).

Changes needed before the next draft, not made here:

- Add the biomarker-by-time specifications (`E2cr2_bmt`, `E7cr2_bmt`, `E7null_cr2_bmt`, `F1A0_bmt`, optionally quadratic) to `simulation-core.R`. Add the corresponding block to `36-run-referee-variants.R` (split plan, Section 3.1, step 6).
- Give `30-check-prognostic-confound-pd.R` a `--params` switch so that the placebo-response contrast can be recomputed under the published SDs.
- Give `37-ladder-tables-figures.R --scope all` a published-set input path if paper 15 is to carry ladder-style figures. Its trend figure (`02-ladder-fig4-trend.png`) was built from historical data and is not used here.

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A Absorption of a constant shift

Let X be the fixed-effects design matrix of any analysis here. It contains the biomarker main effect B_i^c. The constant shift adds $b_{TV}\sigma_{TV}z_i$ to every outcome of participant i. Since z_i is an affine function of B_i^c, the shift vector lies in the column space of X. For fixed variance parameters the generalized least-squares estimate then changes only in the coefficients of the intercept and B_i^c. The residuals are unchanged. REML and ML estimates of the variance parameters depend on the data only through those residuals, so they are unchanged too. AIC selection compares fits whose likelihoods are unchanged, so it selects the same half-life. CR2 standard errors depend on the residuals and the design only. The interaction estimate, its standard error and its p-value are therefore identical, replicate by replicate. The stored data confirm it (Section 3.2).

B Draft consistency checks

The prose of this draft makes 29 qualitative statements that are checked against the stored data when it renders. All of them hold.